Tasty Talks: An Engaging Social Platform Designed for Food Enthusiasts, Chefs, and Interactive Recipe Communities

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology

in The Department of CSE

FULL STACK APPLICATION DEVELOPMENT
24SDCS02A

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Introduction

In the modern digital world, social networking platforms play a crucial role in connecting people with similar interests across the globe. Among the many thriving online communities, the food and culinary domain has gained significant popularity. Millions of users actively search for recipes, cooking techniques, restaurant reviews, and healthy meal ideas through various digital platforms. However, most existing platforms are either general social media applications or simple recipe-sharing websites that lack focused interaction, personalization, and community-driven engagement. This creates a need for a dedicated, interactive social platform specifically designed for food enthusiasts, chefs, and cooking communities.

The project **"Tasty Talks"** falls under the domain of Full Stack Web Application Development and Social Networking Systems. It aims to design and develop a dynamic, user-friendly, and scalable web-based platform where users can share recipes, post food images, exchange cooking ideas, and engage in meaningful discussions related to culinary arts. The system integrates both frontend and backend technologies to provide a seamless user experience while ensuring secure data management and efficient performance.

Large-scale web applications like social networking platforms consist of multiple interconnected subsystems. These systems can be logically divided into modules such as user authentication and authorization, content management, database handling, profile management, and communication interfaces. By partitioning the application into these smaller modules, development becomes more structured, maintainable, and scalable. Tasty Talks follows this modular architecture to ensure flexibility and future enhancements.

The platform enables users to create personal profiles, upload and manage recipes, like and comment on posts, and interact with other members in the community. The backend system manages data storage, retrieval, and security using a structured database system. The frontend interface is designed to be responsive and interactive, ensuring compatibility across various devices such as desktops, tablets, and smartphones. Security mechanisms such as login authentication and session management help protect user data and maintain platform integrity.

One of the major goals of this project is to build an engaging digital ecosystem where food lovers can not only access recipes but also contribute their own ideas and learn from others. Unlike traditional recipe websites, Tasty Talks emphasizes social interaction and community participation. Features such as comments, likes, and user-generated content encourage collaboration and knowledge sharing.

From a technical perspective, this project demonstrates the practical implementation of full stack technologies by integrating client-side development, server-side logic, and database management into a unified system. It showcases how modern web technologies can be combined to develop scalable, secure, and interactive applications tailored to specific domains.

Overall, Tasty Talks represents a specialized social networking solution in the culinary field, addressing the need for an organized, community-driven, and interactive platform for food enthusiasts. The project highlights the importance of modular system design, efficient data handling, and responsive user interfaces in building modern web applications.

Literature Review/ Application Survey

Paper1: Multimodal Recipe Recommendation with Heterogeneous Graph Neural Network

Ruiqi Ouyang, Haodong Huang, Weihua Ou, and Qilong Liu

Ouyang et al. address the limitations of traditional recipe recommendation systems by proposing a Multimodal Heterogeneous Graph Neural Network Recipe Recommendation (MHGRR) architecture. The study begins by identifying a key problem in existing systems: most recipe recommendation approaches rely only on textual information, typically ingredient lists or recipe descriptions. However, recipes are inherently multimodal, containing both text and images, and sometimes even audio. The authors argue that relying on a single modality limits the expressive power of recommendation systems because images convey rich semantic information such as food color, texture, presentation, and visual appeal, which cannot be fully captured through text alone. Since users often select recipes first based on images, then review ingredients and cooking steps, incorporating visual features becomes essential for improving personalization accuracy.

To address this issue, the authors propose a multimodal fusion approach that carefully handles the difference in feature granularity between text and images. Text features, particularly ingredient information, are usually represented in binary form (0 or 1 indicating presence or absence of ingredients), whereas image features consist of pixel values ranging from 0 to 255. This difference in distribution makes direct fusion ineffective. Therefore, the model uses an embedding layer to transform ingredient text into dense vector representations, and applies a shallow Convolutional Neural Network (CNN) to extract low-level visual features from images. By doing so, both modalities are converted into compatible feature representations, allowing meaningful feature fusion.

Beyond multimodal fusion, the paper highlights another important challenge: modelling the complex relationship between users and recipes. User preferences are influenced by personal taste, dietary habits, and historical interactions, making the user–recipe relationship highly complex. Traditional collaborative filtering methods focus mainly on similarity between users or items and fail to capture deeper relational structures. To overcome this, the authors construct a heterogeneous graph consisting of user nodes and recipe nodes connected through interaction history (ratings). They employ a Graph Neural Network (GNN) architecture based on Graph SAGE, which performs neighbour sampling to aggregate information from connected nodes. This allows the model to effectively learn both user features and recipe features while capturing complex interaction patterns. The heterogeneous design helps eliminate structural differences between user nodes and recipe nodes and enables better relational learning.

The study formulates the recommendation problem mathematically by defining a graph G composed of user–recipe interactions, where each interaction is represented as binary (1 if rated, 0 otherwise). The model integrates three key inputs: ingredient text, recipe images, and user rating history. These inputs are processed through embedding layers, CNNs, and a two-layer Graph SAGE network to generate final feature representations. A prediction score is then produced to estimate user preference for a recipe.

Paper2: Food Recipe Recommendation System with Content Based Filtering and Collaborative Filtering Methods

Ni Putu Triska Widiantari, I Made Agus Dwi Suarjaya, and Ni Kadek Dwi Rusjayanthi

Widiantari et al. developed a mobile-based recipe recommendation system aimed at encouraging healthier eating habits by reducing fast-food consumption. The study begins with the observation that frequent fast-food intake increases the risk of serious health problems, including obesity and lifestyle-related diseases. Although many healthy recipes are available online, the overwhelming variety of options makes it difficult for users to choose meals that match their personal preferences and dietary habits. To address this issue, the researchers designed an Android-based recommendation system capable of suggesting suitable recipes based on user behavior and contextual similarity.

The system integrates two primary recommendation techniques: content-based filtering and collaborative filtering. In the content-based filtering approach, recipe similarity is calculated using cosine similarity, which measures the closeness between textual representations of recipes and user preferences. Features such as ingredients and descriptions are transformed into vector representations, and similarity scores are computed to determine how closely a recipe matches user interests. For example, the recipe titled *"chocolate cake that you can easily make at home"* achieved the highest similarity score of 0.41 in their experiments, demonstrating the effectiveness of text-based similarity in identifying relevant food recommendations.

In addition to content-based filtering, the study incorporates Neural Collaborative Filtering (NCF) to model user interactions more effectively. Unlike traditional collaborative filtering methods that rely heavily on explicit ratings, this system uses implicit feedback, such as user comments on recipes, eliminating the need for direct rating inputs. By analysing historical user interactions, the model learns patterns of preference and generates personalized recommendations. The evaluation results show consistent performance improvements after training, with MAP@10 increasing from 0.705 to 0.767 and NDCG@10 improving from 0.78 to 0.83 over 10 training epochs. These results indicate that the neural collaborative approach enhances ranking quality and recommendation accuracy over time.

The system outputs recommendations in the form of the Top-10 food recipes, evaluated using ranking metrics such as NDCG@k (Normalized Discounted Cumulative Gain) and Hit-Ratio@k, which measure the relevance and ranking effectiveness of recommended items. The findings confirm that content-based filtering provides meaningful similarity-based recommendations, while neural collaborative filtering further improves personalization by leveraging user interaction patterns.

Overall, this research demonstrates that combining content-based similarity techniques with neural collaborative filtering can effectively generate personalized food recommendations on mobile platforms. It highlights the importance of using both textual similarity and behavioral data to enhance recommendation quality. For platforms like Tasty Talks, this study provides valuable insights into integrating hybrid recommendation models to improve user engagement, personalization, and healthier food selection support.

Paper3: A Heterogeneous Graph Learning Model for Recipe Recommendation

Yijun Tian, Chuxu Zhang, Zhichun Guo, Chao Huang, Ronald Metoyer, Nitesh V. Chawla

Tian et al. address the limitations of traditional recipe recommendation systems by proposing RecipeRec, a heterogeneous graph learning model designed to capture complex relational structures among users, recipes, and ingredients. Most existing recommendation systems rely primarily on either content-based filtering, which focuses on recipe features such as ingredients and descriptions, or collaborative filtering, which models user–item interactions. However, these approaches often ignore higher-order collaborative signals and the rich relational structure that exists among users, recipes, and food components. To overcome this limitation, the authors formalize recipe recommendation as a graph learning problem.

The study introduces a large-scale User–Recipe–Ingredient Graph (URI-Graph), which models three types of nodes: users, recipes, and ingredients. By structuring the data as a heterogeneous graph, the model captures not only direct user–recipe interactions but also indirect relationships, such as shared ingredients between recipes and preference similarities between users. This graph-based modelling allows the system to leverage deeper relational information that traditional recommendation methods fail to consider.

To effectively learn from this heterogeneous graph, the authors propose the Recipe Rec model, which integrates a heterogeneous graph neural network (HGNN) with hierarchical attention mechanisms. The hierarchical attention enables the model to selectively aggregate information from different types of neighbouring nodes, ensuring that important relational signals are emphasized while less relevant ones are minimized. Additionally, an ingredient set transformer is introduced to encode ingredient-level information more effectively, enhancing the representation of recipe content within the graph framework.

A key innovation in this study is the incorporation of graph contrastive learning. The authors design a graph contrastive augmentation strategy that generates different graph views in a self-supervised manner. By contrasting these augmented representations, the model learns more informative and robust graph embeddings. The final model is optimized using a joint objective function, combining both recommendation loss and contrastive learning loss, which improves feature extraction and enhances overall recommendation accuracy.

Extensive experiments conducted on large-scale datasets demonstrate that Recipe Rec significantly outperforms state-of-the-art baseline models in recipe recommendation tasks. The results confirm that heterogeneous graph learning, combined with contrastive learning techniques, is highly effective in capturing collaborative and content signals simultaneously.

For platforms like Tasty Talks, this research provides valuable insights into leveraging graph-based architectures to enhance personalization. By modelling relationships among users, recipes, and ingredients, Tasty Talks could implement advanced recommendation modules that consider user interactions, ingredient similarity, and community-driven preferences, leading to more accurate and context-aware food suggestions.

Paper4: Personalized, Health-Aware Recipe Recommendation, An Ensemble Topic Modelling Based Approach

Mansura A. Khan, Ellen Rushe, Barry Smyth, David Coyl

Khan et al. focus on the critical relationship between food choices and long-term health, emphasizing that dietary decisions are deeply personal, complex, and influenced by multiple factors such as taste preferences, nutritional needs, lifestyle, demographics, and cost considerations. The study highlights that recommender systems (RS) can play a significant role in guiding users toward healthier eating habits by offering personalized and informed food recommendations. However, a major challenge in building effective food recommender systems lies in intelligent user modelling — accurately understanding and representing user preferences in a dynamic and efficient way.

To address this challenge, the authors propose an approach based on Ensemble Topic Modelling (EnsTM) for feature identification and user modelling. Topic modelling techniques are used to uncover hidden thematic structures in large textual datasets, such as recipe descriptions and ingredient lists. Instead of relying on a single topic model, the researchers combine multiple topic models into an ensemble framework, enabling the system to capture diverse and richer latent features. This ensemble approach improves representation of important attributes such as nutritional content, ingredient relationships, taste profiles, and contextual aspects.

A key contribution of the paper is the development of a reduced data representation format and a smart user modelling strategy that allows rapid and interactive preference capture. This makes the system effective even in cold-start scenarios, where little or no prior user interaction data is available. The authors propose three variants of their recommendation system: two purely EnsTM-based models and one hybrid EnsTM-based recommender. These approaches were evaluated through a user study involving 48 participants and a large-scale real-world dataset containing 230,876 recipes.

Experimental results show that EnsTM-based recommenders significantly outperform conventional content-based (CB) recommendation approaches. The system demonstrates better personalization, improved coverage, and the ability to implicitly understand users' food practices. Importantly, the model integrates nutritional preferences, allowing users to explore recipes under different nutritional categories and promoting healthier food decisions. The analysis also reveals correlations between certain extracted features and healthier lifestyle patterns, indicating the broader health impact potential of such intelligent recommender systems.

Overall, this research underscores the importance of integrating health-awareness, multi-domain preference modelling, and advanced topic modelling techniques in recipe recommendation systems. For platforms like Tasty Talks, this study suggests future enhancements such as incorporating nutritional filtering, diet-based recommendations, and intelligent user preference modelling to support users with specific health goals while maintaining personalization and engagement.

Paper5: A Smart Platform for Food Enthusiasts Driven by AI for Recipe Sharing and Discovery

Elangovan T, Mukesh M, Jeevan M S, Satheesh P

This study proposes an AI-driven smart recipe recommendation platform designed to enhance recipe discovery and personalization through advanced Natural Language Processing (NLP) and real-time user interaction. The authors emphasize that food plays a fundamental role in human life and cultural connection, and with the rapid growth of digital platforms, users increasingly rely on online systems to discover and share recipes. However, most existing recipe platforms operate on static databases with traditional keyword-based search mechanisms and limited personalization, making it difficult for users to find recipes that match their specific preferences, dietary needs, and contextual requirements.

To overcome these limitations, the paper introduces an intelligent recipe recommendation system built using BERT (Bidirectional Encoder Representations from Transformers). Unlike conventional search models that rely on simple keyword matching, BERT enables bidirectional contextual understanding of user queries. This allows the system to interpret not only ingredient names but also cooking methods, semantic context, and user intent. For example, when a user searches for recipes containing specific ingredients, the system can understand how those ingredients are used within different culinary contexts and suggest recipes accordingly. This improves semantic organization and relevance compared to traditional search approaches.

A significant contribution of the proposed system is its dynamic adaptability feature, which integrates real-time user feedback such as ratings and comments into the recommendation process. The model continuously updates its predictions based on user interactions, allowing it to refine personalization over time. This interactive learning mechanism enhances user engagement and satisfaction by tailoring suggestions according to individual tastes, dietary restrictions, and cooking preferences.

The study also highlights scalability as a key feature of modern AI-based recipe platforms. As datasets grow larger and include more complex ingredient combinations, the system must efficiently handle high-dimensional textual and contextual data. The authors demonstrate that their BERT-based approach achieves significantly improved recommendation accuracy and user engagement compared to traditional models. The research contributes to the emerging domain of intelligent recipe platforms by combining contextual language understanding, personalization, and interactive feedback into a unified AI framework.

Overall, this work emphasizes the importance of contextual NLP models and adaptive learning strategies in modern recipe recommendation systems. For platforms like TastyTalks, this research supports the integration of AI-powered ingredient-based search, contextual query understanding, and feedback-driven personalization to create a more interactive and intelligent food-sharing ecosystem.

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